

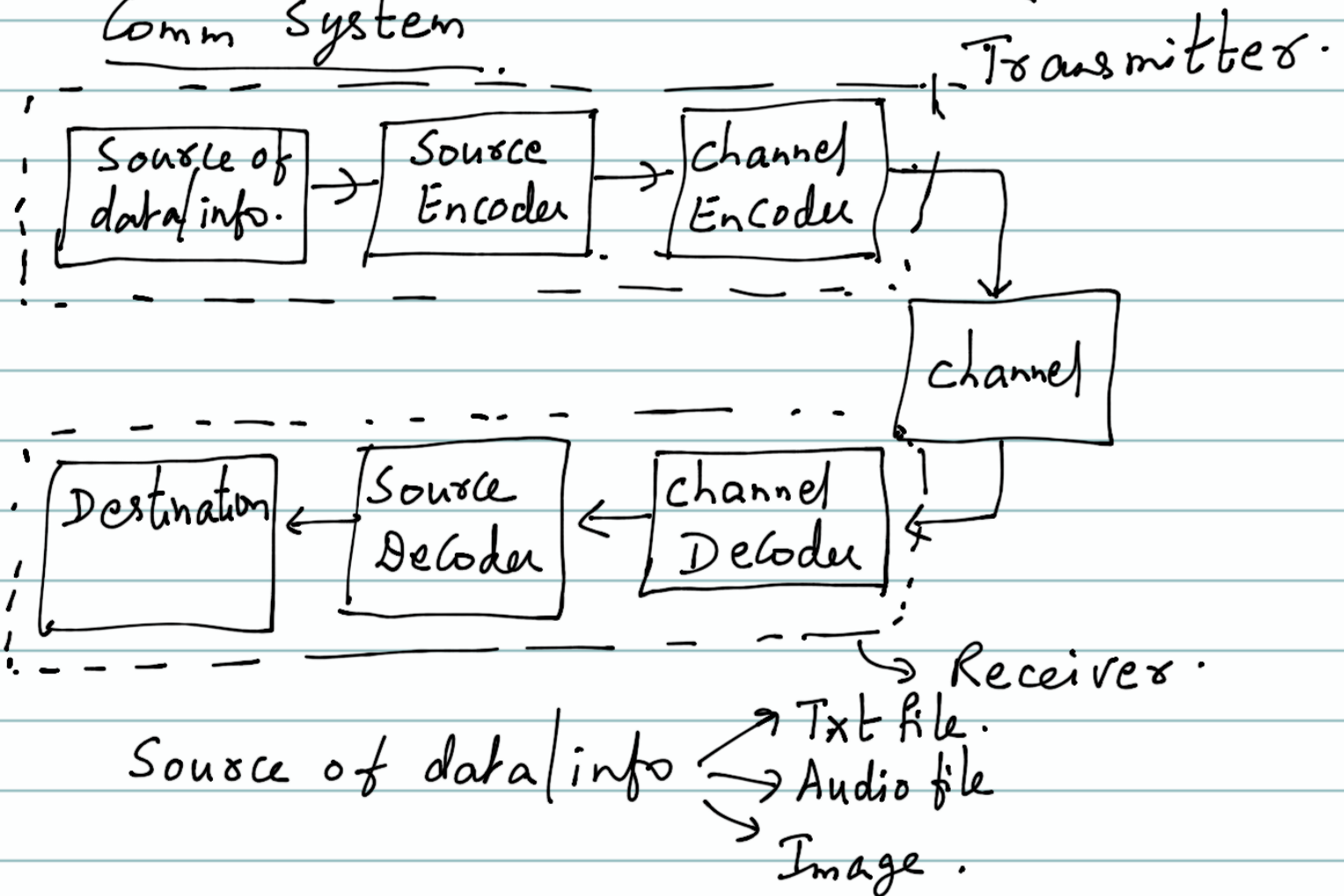
# Motivation

## Communication System

Examples:- ① Telephones, mobiles, TV, mails (Comm in space).

② CD, DVDs, flash drives, (Memory info).  
(Comm in time)

## Block diagram / (back bone) of a Comm System



Source Encoder:- Compresses data and <sup>sequence of source symbols</sup> converts into bits. (bit stream) → Removes Redundancy

Example:- Zip a bunch of files.  
There is natural redundancy present in the data.

Source Decoder Converts the bit stream back into source symbols.

We will not deal with Source encoder & decoder more in the course.  
( Zip → Lempel Ziv 77/78 algorithm  
Compression of images → Huffman algorithm  
(JPEG) )

Channel Encoder:- Takes as input the bit stream → introduces redundancy in a controlled fashion. →

Why are adding this redundancy?

Because there is channel which is coming ahead which will corrupt your message.

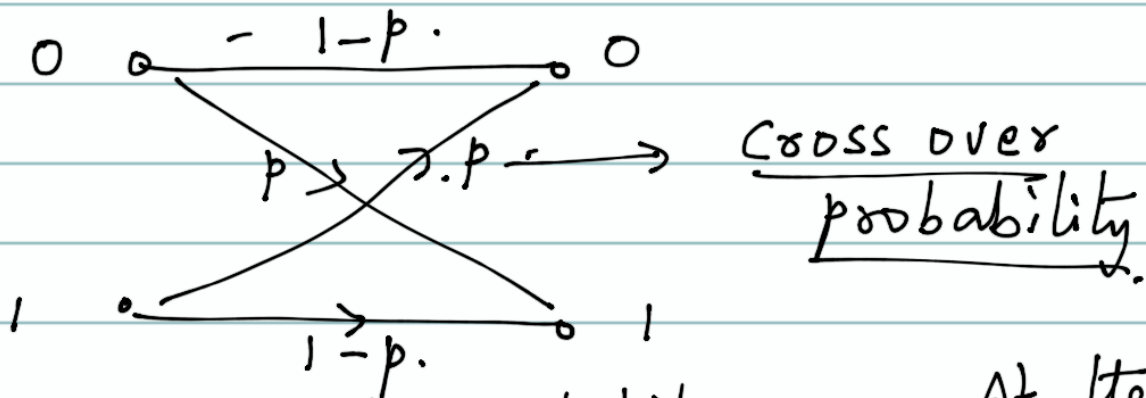
Desirable → To be able to correctly/reliably receive the message.

Channel Decoder:- Receives the corrupted Codeword and tries to guess whatever was sent by the transmitter

Examples of Channel Models



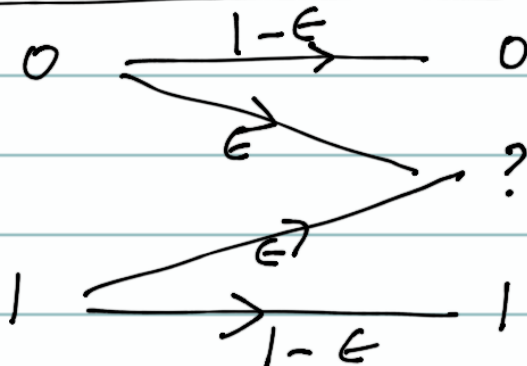
Binary Symmetric channel



0010  $\xrightarrow{\text{Flips each bit independently with prob } p.}$

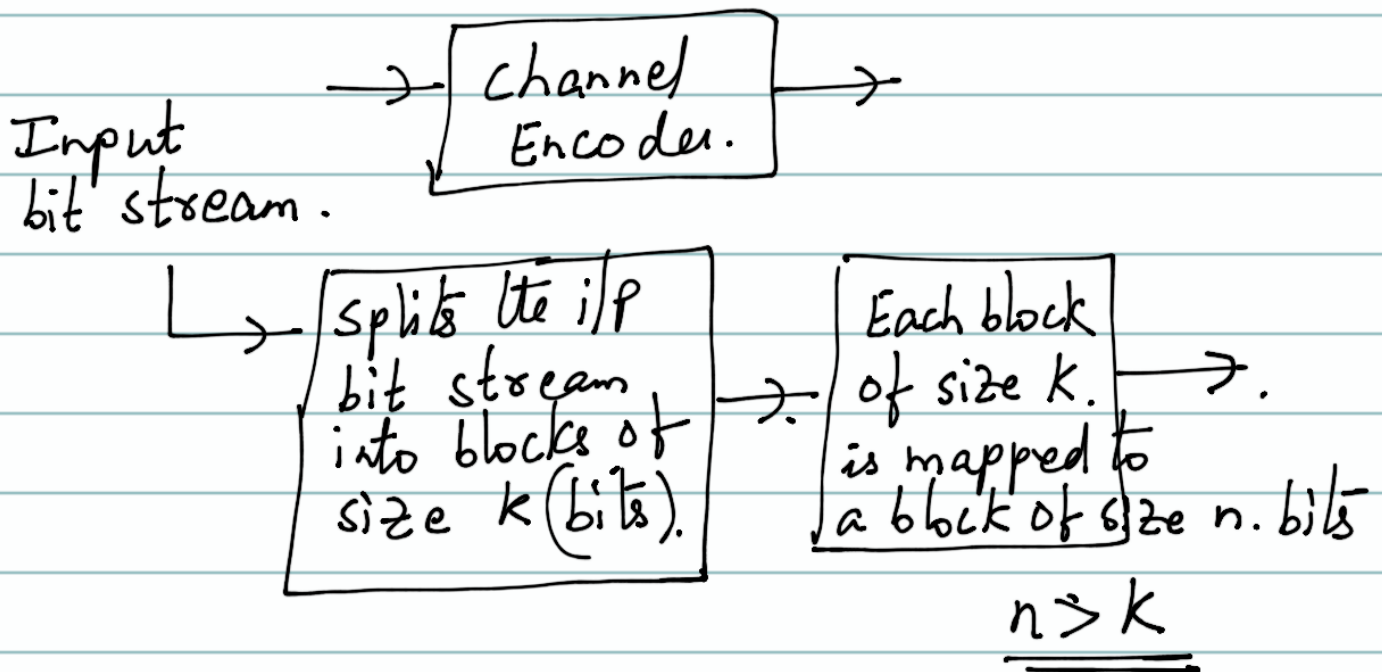
At the receiver, you can get any of the 16 possible combinations.

Binary Erasure channel



0010  $\rightarrow$  16 outputs possible

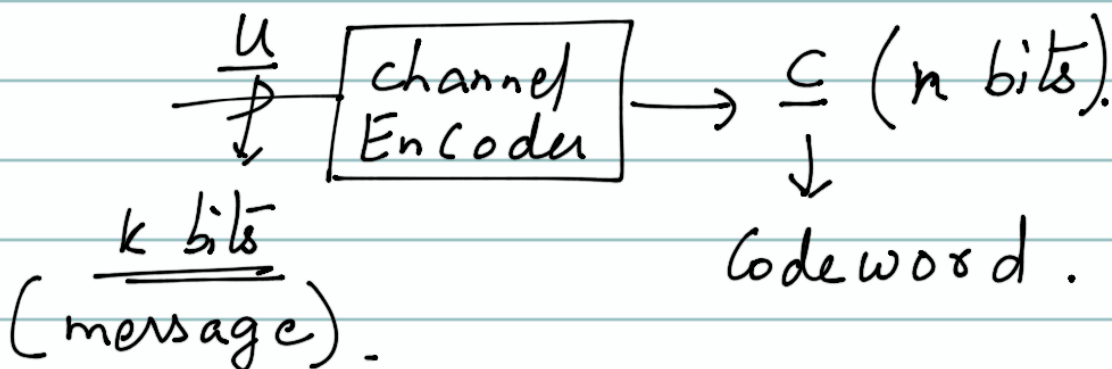
# channel Encoder



00100111

Simple example of a channel encoder is a repetition code — 3-fold repetition  
 $k=1, n=3$

00100111 → 000 000 111 000 000  
 8 bits → 24 bits. 111 111 111



$k$  bit message  $\underline{u} \rightarrow 2^k$  distinct messages are possible.

How many distinct codewords should the encoder map to?

No. of distinct codewords =  $2^k$ .

Channel Encoder map is a one-one function.

$$\begin{array}{l} 0 \searrow \\ 1 \nearrow \end{array} 000.$$

A code is the collection of all possible codewords. (as a set)

$(000, 111)$

$0 \rightarrow 000$

$1 \rightarrow 111$

Let's say the received vector is 110.  $\rightarrow$  What's your guess of transmitted vector? (depends on the flipping probability)

Maximum likelihood decoding  $\rightarrow$  Best channel decoder for any channel



## Probability of error calculations assuming your channel is BSC.

Assume  $p = 0.1$ .

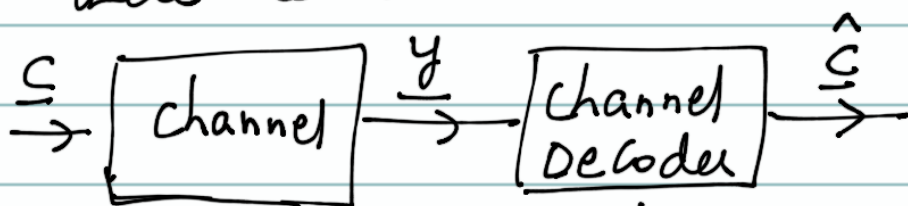
### Decoding Rule

Received Vector  
(Corrupted)  $\longrightarrow$  Codewords.  
(Messages are in one-to-one map with the Codewords).

### 3-fold repetition Code

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| 000 | 001 | 010 | 011 | 100 | 101 | 110 | 111 |
| ↓   | ↓   | ↓   | ↓   | ↓   | ↓   | ↓   | ↓   |
| 000 | 000 | 000 | 111 | 000 | 111 | 111 | 111 |

Even when you make your best guess,  
there is a chance that you are in error.



Error Event:  $\{c \neq \hat{c}\}$ .  
 $P(c \neq \hat{c})$ .

$$\begin{aligned}
 P(\underline{c} \neq \hat{\underline{c}}) &= P(\underline{c} \neq \hat{\underline{c}} \mid \underline{c} = 000) \cdot P(\underline{c} = 000) + P(\underline{c} \neq \hat{\underline{c}} \mid \underline{c} = 111) \cdot P(\underline{c} = 111). \quad \left( \begin{array}{l} \text{Using total} \\ \text{prob. theorem} \end{array} \right)
 \end{aligned}$$

$$P(\underline{c} = 000) = P(\underline{u} = 0) = \frac{1}{2} = P(\underline{c} = 111) = P(\underline{u} = 1)$$

$$\begin{aligned}
 &P(\underline{c} \neq \hat{\underline{c}} \mid \underline{c} = 000) \\
 &= P(\underline{y} = 110 \mid \underline{c} = 000) + P(\underline{y} = 011 \mid \underline{c} = 000) \\
 &\quad + P(\underline{y} = 101 \mid \underline{c} = 000) \\
 &\quad + P(\underline{y} = 111 \mid \underline{c} = 000) \\
 &\approx p \cdot p \cdot (1-p) + p \cdot p \cdot (1-p) \\
 &\quad + p \cdot p \cdot (1-p) + p \cdot p \cdot p \\
 &\approx p^3 + 3p^2(1-p).
 \end{aligned}$$

$$\begin{aligned}
 &P(\underline{c} \neq \hat{\underline{c}} \mid \underline{c} = 111) \\
 &= p^3 + 3p^2(1-p)
 \end{aligned}$$

$$P(\underline{c} \neq \hat{\underline{c}}) = p^3 + 3p^2(1-p).$$

Uncoded (Not using channel code)

$$\rightarrow P(\underline{c} \neq \hat{\underline{c}}) = 0.1$$

3-fold repetition

$$\rightarrow P(\underline{c} \neq \hat{\underline{c}}) = 3p^2(1-p) + p^3 \\ = 0.028.$$

Prob. of error decreases by employing a nontrivial channel code.

Metric for reliability is probability of error.

$$5\text{-fold} \rightarrow p_e^5$$

$$7\text{-fold} \rightarrow p_e^7$$

$$p_e^1 > p_e^3 > p_e^5 > p_e^7$$

As a system designer, for a target prob. of error, choose the repetition factor.